

Database Management System

UNIT-1

1. Introduction to database systems:

Database: Database is a collection of inter-related data which helps in efficient retrieval, insertion and deletion of data from database and organizes the data in the form of tables, views, schemas, reports etc. For Example, university database organizes the data about students, faculty, and admin staff etc. which helps in efficient retrieval, insertion and deletion of data from it.

A database is a collection of related data which represents some aspect of the real world. A database system is designed to be built and populated with data for a certain task.

Let us see a simple example of a university database. This database is maintaining information concerning students, courses, and grades in a university environment. The database is organized as five files:

- The STUDENT file stores data of each student
- The COURSE file stores contain data on each course.
- The SECTION stores the information about sections in a particular course.
- The GRADE file stores the grades which students receive in the various sections
- The TUTOR file contains information about each professor.

DBMS:

Database Management System (DBMS) is a software for storing and retrieving users' data while considering appropriate security measures. It consists of a group of programs which manipulate the database. The DBMS accepts the request for data from an application and instructs the operating system to provide the specific data. In large systems, a DBMS helps users and other third-party software to store and retrieve data.

The software which is used to manage database is called Database Management System (DBMS). For Example, MySQL, Oracle etc. are popular commercial DBMS used in different applications.

Popular DBMS Software:

Here, is the list of some popular DBMS system:

- MySQL
- Microsoft Access
- Oracle
- PostgreSQL
- dBASE

- FoxPro
- SQLite
- IBM DB2
- LibreOffice Base
- MariaDB
- Microsoft SQL Server etc.

Application of DBMS

Sector	Use of DBMS
Banking	For customer information, account activities, payments, deposits, loans, etc.
Airlines	For reservations and schedule information.
Universities	For student information, course registrations, colleges and grades.
Telecommunication	It helps to keep call records, monthly bills, maintaining balances, etc.
Finance	For storing information about stock, sales, and purchases of financial instruments like stocks and bonds.
Sales	Use for storing customer, product & sales information.
Manufacturing	It is used for the management of supply chain and for tracking production of items. Inventories status in warehouses.
HR Management	For information about employees, salaries, payroll, deduction, generation of paychecks, etc.

DBMS allows users the following tasks:

Data Definition: It helps in creation, modification and removal of definitions that define the organization of data in database.

Data Updation: It helps in insertion, modification and deletion of the actual data in the database.

Data Retrieval: It helps in retrieval of data from the database which can be used by applications for various purposes.

User Administration: It helps in registering and monitoring users, enforcing data security, monitoring performance, maintaining data integrity, dealing with concurrency control and recovering information corrupted by unexpected failure.

Types of databases:

Depending upon the usage requirements, there are following types of databases available in the market

There are various types of databases used for storing different varieties of data

- Centralized database.
- Distributed database.
- Personal database.
- End-user database.
- Commercial database.
- NoSQL database.
- Operational database.
- Relational database.
- Cloud database.
- Object-oriented database.
- Graph database.

2. Difference between File System and DBMS:

What is File System?

A File Management system is a DBMS that allows access to single files or tables at a time. In a File System, data is directly stored in set of files. It contains flat files that have no relation to other files (when only one table is stored in single file, then this file is known as flat file).

File system is basically a way of arranging the files in a storage medium like hard disk. File system organizes the files and helps in retrieval of files when they are required. File systems consists of different files which are grouped into directories. The directories further contain other folders and files. File system performs basic operations like management, file naming, giving access rules etc.

Example: NTFS (New Technology File System), EXT (Extended File System).

What is DBMS?

Database Management System is basically a software that manages the collection of related data. It is used for storing data and retrieving the data effectively when it is needed. It also provides proper security measures for protecting the data from unauthorized access. In Database Management System the data can be fetched by SQL queries and relational algebra. It also provides mechanisms for data recovery and data backup.

Example:

Oracle, MySQL, MS SQL server.

Advantages of DBMS over File system:

Data redundancy and inconsistency:

Redundancy is the concept of repetition of data i.e. each data may have more than a single copy. The file system cannot control redundancy of data as each user defines and maintains the needed files for a specific application to run. There may be a possibility that two users are maintaining same files data for different applications. Hence changes made by one user does not reflect in files used by second users, which leads to inconsistency of data. Whereas DBMS controls redundancy by maintaining a single repository of data that is defined once and is accessed by many users. As there is no or less redundancy, data remains consistent.

Data sharing:

File system does not allow sharing of data or sharing is too complex. Whereas in DBMS, data can be shared easily due to centralized system.

Data concurrency:

Concurrent access to data means more than one user is accessing the same data at the same time. Anomalies occur when changes made by one user gets lost because of changes made by other user. File system does not provide any procedure to stop anomalies. Whereas DBMS provides a locking system to stop anomalies to occur.

Data searching:

For every search operation performed on file system, a different application program has to be written. While DBMS provides inbuilt searching operations. User only have to write a small query to retrieve data from database.

Data integrity:

There may be cases when some constraints need to be applied on the data before inserting it in database. The file system does not provide any procedure to check these constraints automatically. Whereas DBMS maintains data integrity by enforcing user defined constraints on data by itself.

System crashing:

In some cases, systems might have crashes due to various reasons. It is a bane in case of file systems because once the system crashes, there will be no recovery of the data that's been lost. A DBMS will have the recovery manager which retrieves the data making it another advantage over file-systems.

Data security:

A file system provides a password mechanism to protect the database but how longer can the password be protected? No one can guarantee that. This doesn't happen in the case of DBMS. DBMS has specialized features that help provide shielding to its data.

S.NO.	File System	DBMS
1.	File system is a software that manages and organizes the files in a storage medium within a computer.	DBMS is a software for managing the database.
2.	Redundant data can be present in a file system.	In DBMS there is no redundant data.
3.	It doesn't provide backup and recovery of data if it is lost.	It provides backup and recovery of data even if it is lost.
4.	There is no efficient query processing in file system.	Efficient query processing is there in DBMS.
5.	There is less data consistency in file system.	There is more data consistency because of the process of normalization.
6.	It is less complex as compared to DBMS.	It has more complexity in handling as compared to file system.
7.	File systems provide less security in comparison to DBMS.	DBMS has more security mechanisms as compared to file system.
8.	It is less expensive than DBMS.	It has a comparatively higher cost than a file system.

Advantages of DBMS.

Due to its centralized nature, the database system can overcome the disadvantages of the file system-based system

1. Data independency:

Application program should not be exposed to details of data representation and storage DBMS provides the abstract view that hides these details.

2. Efficient data access:

DBMS utilizes a variety of sophisticated techniques to store and retrieve data efficiently.

3. Data integrity and security:

Data is accessed through DBMS, it can enforce integrity constraints. E.g.: Inserting salary information for an employee.

4. Data Administration:

When users share data, centralizing the data is an important task, Experience professionals can minimize data redundancy and perform fine tuning which reduces retrieval time.

5. Concurrent access and Crash recovery:

DBMS schedules concurrent access to the data. DBMS protects user from the effects of system failure.

6. Reduced application development time.

DBMS supports important functions that are common to many applications.

3. Database system structure:

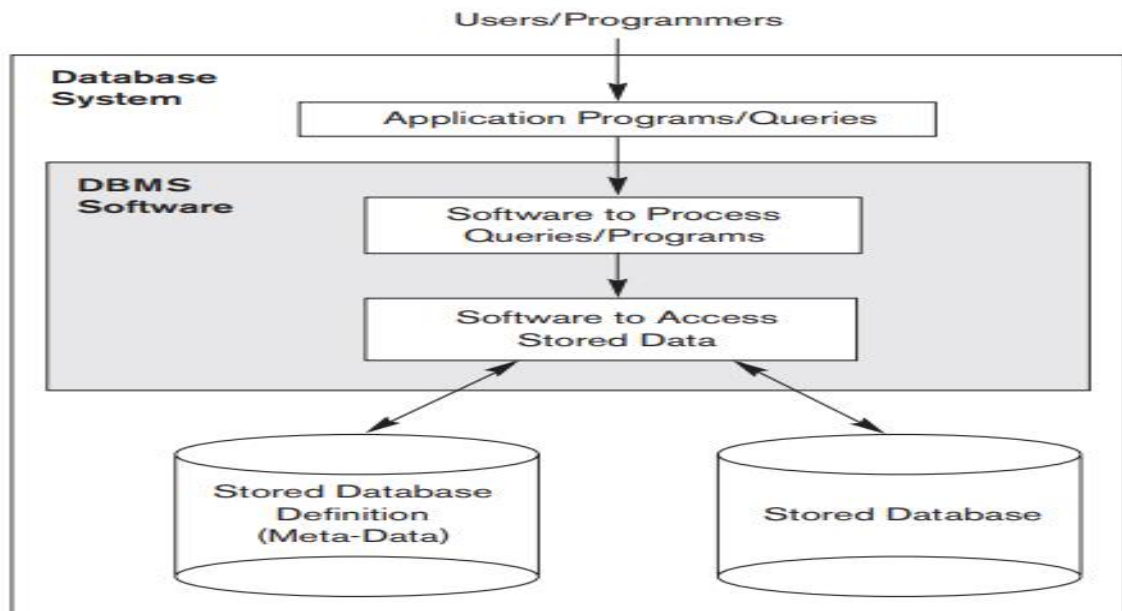
Simplified Database System Environment:

A database management system (DBMS) is a collection of programs that enables users to create and maintain database. The DBMS is a general purpose software system that facilitates the process of defining, constructing, manipulating and sharing databases among various users and applications. Defining a database specifying the database involves specifying the data types, constraints and structures of the data to be stored in the database. The descriptive information is also stored in the database in the form database catalog or dictionary; it is called meta-data.

Manipulating the data includes the querying the database to retrieve the specific data.

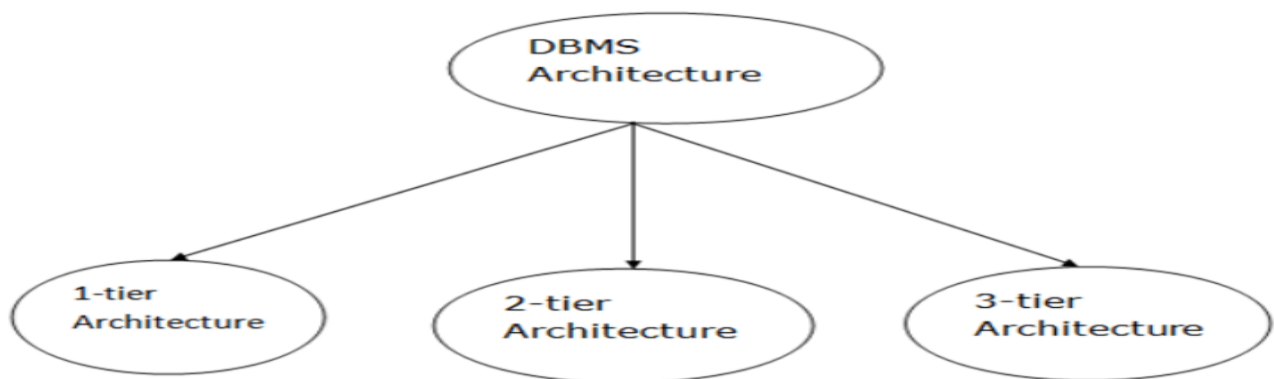
An application program accesses the database by sending the queries or requests for data to DBMS.

The important function provided by the DBMS includes protecting the database and maintain the database.



DBMS Architecture

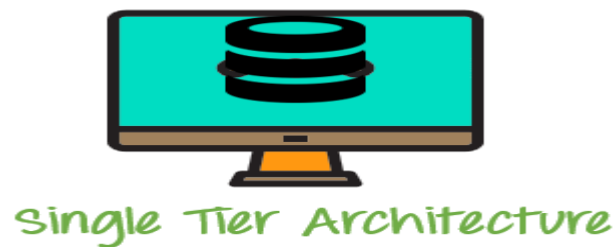
- The DBMS design depends upon its architecture. The basic client/server architecture is used to deal with a large number of PCs, web servers, database servers and other components that are connected with networks.
- The client/server architecture consists of many PCs and a workstation which are connected via the network.
- DBMS architecture depends upon how users are connected to the database to get their request done.



Database architecture can be seen as a single tier or multi-tier. But logically, database architecture is of two types like: **2-tier architecture** and **3-tier architecture**.

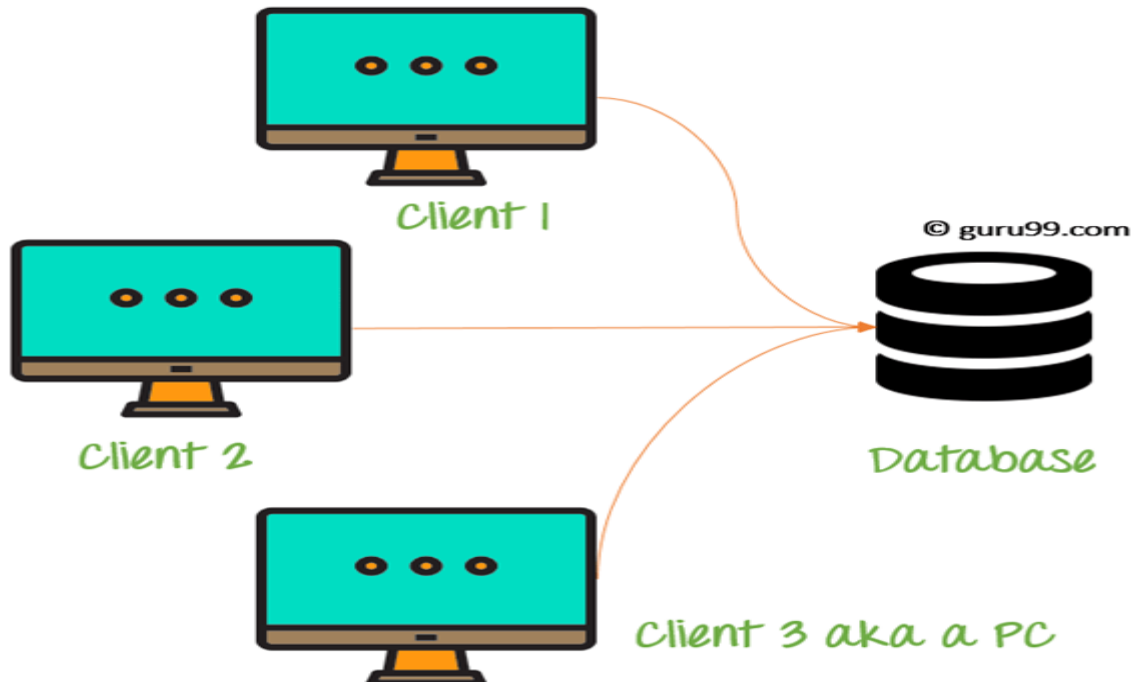
1-Tier Architecture:

1 Tier Architecture in DBMS is the simplest architecture of Database in which the client, server, and Database all reside on the same machine. A simple one tier architecture example would be anytime you install a Database in your system and access it to practice SQL queries. But such architecture is rarely used in production.



2-Tier Architecture:

A **2 Tier Architecture** in DBMS is a Database architecture where the presentation layer runs on a client (PC, Mobile, Tablet, etc.), and data is stored on a server called the second tier. Two tier architecture provides added security to the DBMS as it is not exposed to the end-user directly. It also provides direct and faster communication.



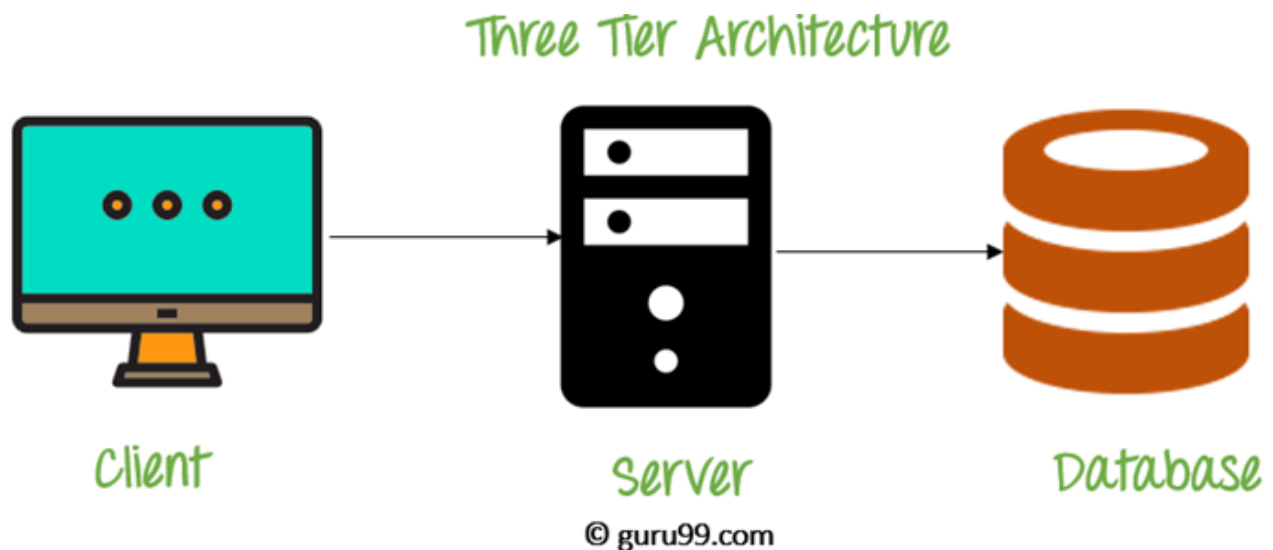
In the above 2 Tier client-server architecture of database management system, we can see that one server is connected with clients 1, 2, and 3.

3-Tier Architecture:

A **3 Tier Architecture** in DBMS is the most popular client server architecture in DBMS in which the development and maintenance of functional processes, logic, data access, data storage, and user interface is done independently as separate modules. Three Tier architecture contains a presentation layer, an application layer, and a database server.

3-Tier database Architecture design is an extension of the 2-tier client-server architecture. A 3-tier architecture has the following layers:

1. Presentation layer (your PC, Tablet, Mobile, etc.)
2. Application layer (server)
3. Database Server



The Application layer resides between the user and the DBMS, which is responsible for communicating the user's request to the DBMS system and send the response from the DBMS to the user. The application layer (business logic layer) also processes functional logic, constraint, and rules before passing data to the user or down to the DBMS.

The goal of Three Tier client-server architecture is:

- To separate the user applications and physical database
- To support DBMS characteristics
- Program-data independence
- Supporting multiple views of the data

Three Tier Architecture Example:

Any large website on the internet, including **guru99.com**.

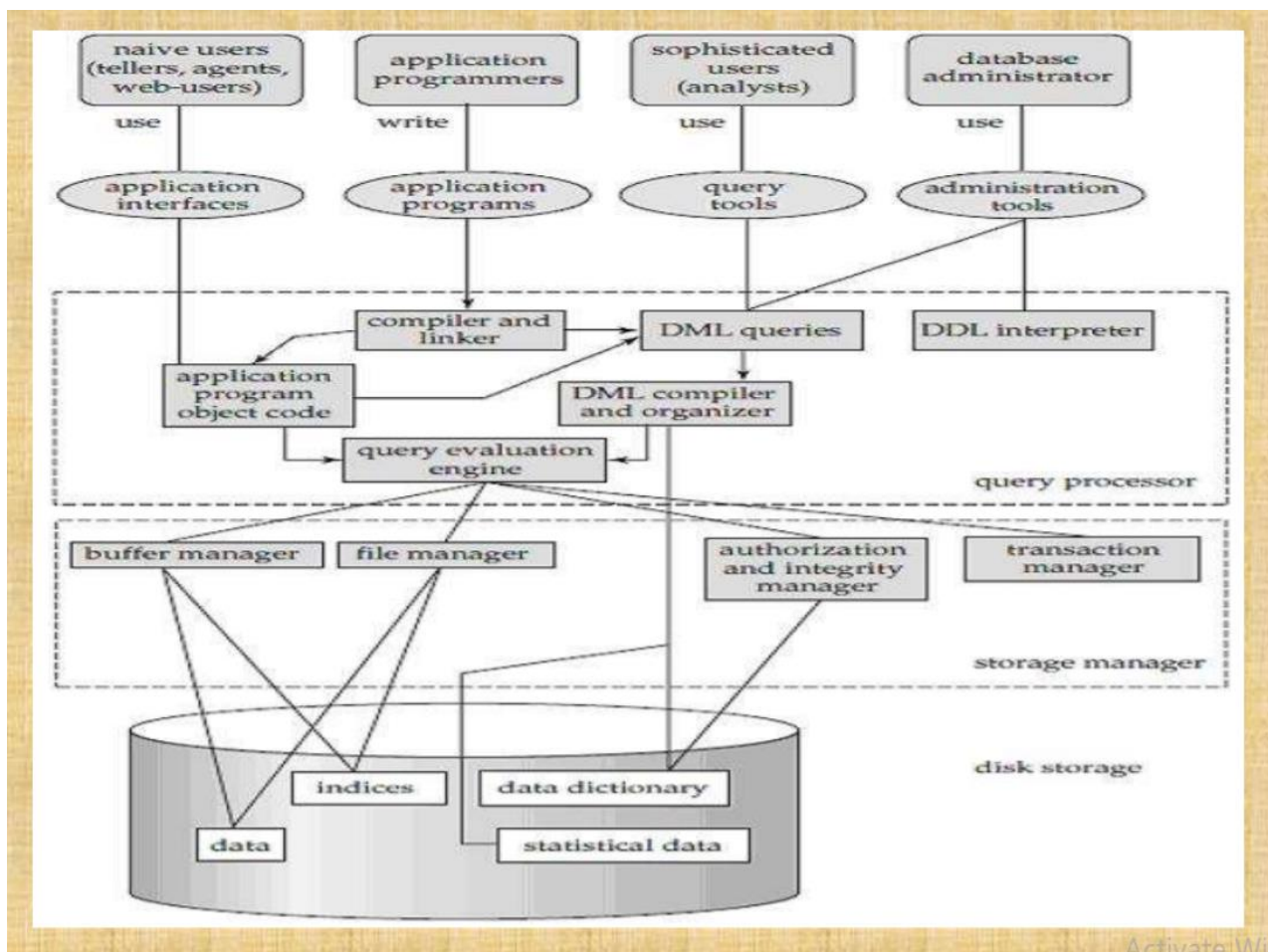
Structure of Database Management System:

Database Management System (DBMS) is a software that allows access to data stored in a database and provides an easy and effective method of –

- Defining the information.
- Storing the information.
- Manipulating the information.
- Protecting the information from system crashes or data theft.
- Differentiating access permissions for different users.

The database system is divided into three components: Query Processor, Storage Manager, and Disk Storage.

A database system is partitioned into modules that deal with each of the responsibilities of the overall system. The functional components of a database system can be broadly divided into the **storage manager** and the **query processor** components. The storage manager is important because databases typically require a large amount of storage space. The query processor is important because it helps the database system simplify and facilitate access to data. It is the job of the database system to translate updates and queries written in a nonprocedural language, at the logical level, into an efficient sequence of operations at the physical level.



Query Processor:

The query processor components include

DDL interpreter, it processes the DDL statements into a set of table containing Meta data (data about data). Which interprets DDL statements and records the definitions in the data dictionary.

DML compiler, which translates DML statements in a query language into an evaluation plan consisting of low-level instructions that the query evaluation engine understands.

Query evaluation engine, which executes low-level instructions generated by the DML compiler.

Storage Manager:

Storage Manager is a program that provides an interface between the data stored in the database and the queries received. It is also known as Database Control System. It maintains the consistency and integrity of the database by applying the constraints and executes the DCL statements. It is responsible for updating, storing, deleting, and retrieving data in the database.

It contains the following components –

- **Authorization Manager**
It ensures role-based access control, i.e.,. Checks whether the particular person is privileged to perform the requested operation or not.
- **Integrity Manager**
It checks the integrity constraints when the database is modified.
- **Transaction Manager**
It controls concurrent access by performing the operations in a scheduled way that it receives the transaction. Thus, it ensures that the database remains in the consistent state before and after the execution of a transaction.
- **File Manager –**
It manages the file space and the data structure used to represent information in the database.
- **Buffer Manager –**
It is responsible for cache memory and the transfer of data between the secondary storage and main memory.

Disk Storage:

It contains the following components –

- **Data Files** –
It stores the data.
- **Data Dictionary** –
It contains the information about the structure of any database object. It is the repository of information that governs the metadata.
- **Indices** –
It provides faster retrieval of data item.

Types of Database Users

1. Database Administrator (DBA):

DBA Stands for Database Administrator.

It is a person or a team, who is responsible for managing the overall database management system. DBA is responsible for:

- Deciding the instances for the database.
- Defining the Schema
- Define Security
- Back-up and Recovery
- Monitoring the performance

2. Database Designers:

- Database designers design the appropriate structure for the database, where we share data.

3. System Analyst:

- **System analyst** analyses the requirements of end users, especially naïve and parametric end users.

4. Application Programmers:

- Application programmers are computer professionals, who write application programs.

5. Naïve Users / Parametric Users:

- Naïve Users are Un-sophisticated users, which has no knowledge of the database. These users are like a layman, which has a little bit of knowledge of the database.
- Naive Users are just to work on developed applications and get the desired result.
- For Example: Railway's ticket booking users are naive users. Or Clerical staff in any bank is a naïve user because they don't have any DBMS knowledge but they still use the database and perform their given task.

6. Sophisticated Users:

- Sophisticated users can be engineers, scientists, business analyst, who are familiar with the database. These users interact with the database but they do not write programs

7. Casual Users / Temporary Users:

- These types of users communicate with the database for a little period of time.

4. Views of data in a database system:

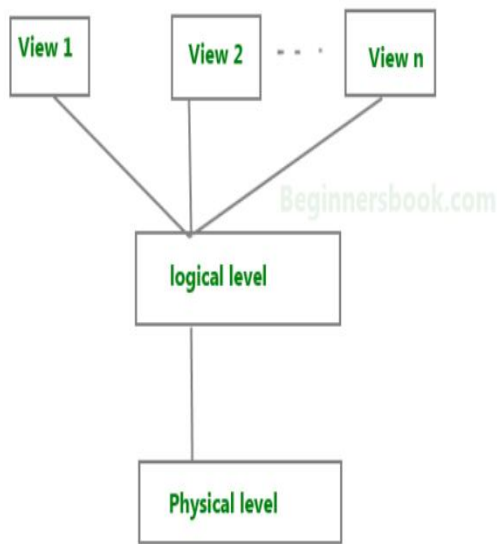
Abstraction is one of the main features of database systems. Hiding irrelevant details from user and providing abstract view of data to users, helps in easy and efficient **user-database** interaction. The view level provides the “**view of data**” to the users and hides the irrelevant details such as data relationship, database schema, constraints, security etc. from the user.

To fully understand the view of data, you must have a basic knowledge of data abstraction and instance & schema..

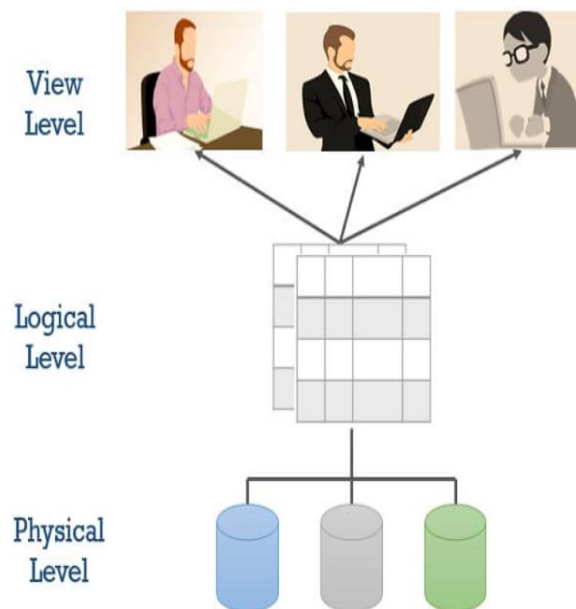
1. Data abstraction
2. Instance and schema

Data Abstraction:

Database systems are made-up of complex data structures. To ease the user interaction with database, the developers hide internal irrelevant details from users. This process of hiding irrelevant details from user is called data abstraction.



Three Levels of data abstraction



We have three levels of abstraction:

Physical level: This is the lowest level of data abstraction. It describes how data is actually stored in database. You can get the complex data structure details at this level.

Logical level: This is the middle level of 3-level data abstraction architecture. It describes what data is stored in database.

View level: Highest level of data abstraction. This level describes the user interaction with database system.

Example: Let's say we are storing customer information in a customer table. At **physical level** these records can be described as blocks of storage (bytes, gigabytes, terabytes etc.) in memory. These details are often hidden from the programmers.

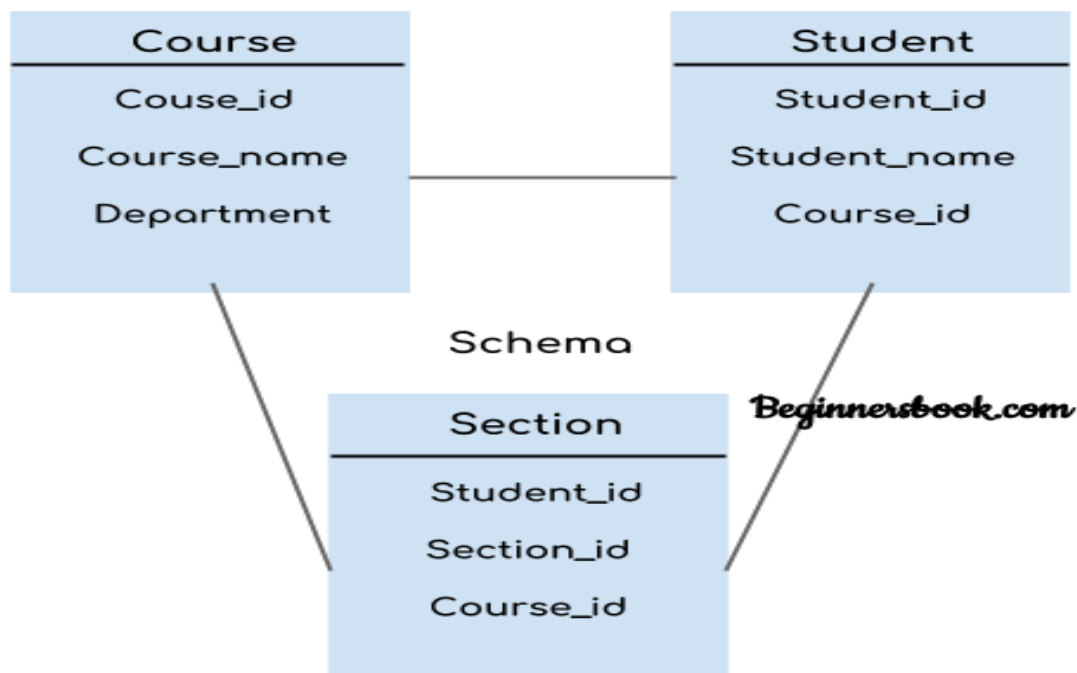
At the **logical level** these records can be described as fields and attributes along with their data types, their relationship among each other can be logically implemented. The programmers generally work at this level because they are aware of such things about database systems.

At view level, user just interact with system with the help of GUI and enter the details at the screen, they are not aware of how the data is stored and what data is stored; such details are hidden from them.

DBMS Schema:

Definition of schema: Design of a database is called the schema. Schema is of three types: Physical schema, logical schema and view schema.

For example: In the following diagram, we have a schema that shows the relationship between three tables: Course, Student and Section. The diagram only shows the design of the database, it doesn't show the data present in those tables. Schema is only a structural view(design) of a database.



The design of a database at physical level is called **physical schema**, how the data stored in blocks of storage is described at this level.

Design of database at logical level is called **logical schema**, programmers and database administrators work at this level, at this level data can be described as certain types of data records gets stored in data structures, however the internal details such as implementation of data structure is hidden at this level (available at physical level).

Design of database at view level is called **view schema**. This generally describes end user interaction with database systems.

DBMS Instance:

Definition of instance: The data stored in database at a particular moment of time is called instance of database. Database schema defines the variable declarations in tables that belong to a particular database; the value of these variables at a moment of time is called the instance of that database.

For example, let's say we have a single table student in the database, today the table has 100 records, and so today the instance of the database has 100 records. Let's say we are going to add another 100 records in this table by tomorrow so the instance of database tomorrow will have 200 records in table. In short, at a particular moment the data stored in database is called the instance that changes over time when we add or delete data from the database.

5. Database Models:

A Database model defines the logical design and structure of a database and defines how data will be stored, accessed and updated in a database management system. While the **Relational Model** is the most widely used database model, there are other models too:

1. Hierarchical Model
2. Network Model
3. Entity-Relationship Model
4. Relational Model
5. Object-Oriented Data Model
6. Object-Relational Data Model
7. Flat Data Model
8. Semi-Structured Data Model
9. Associative Data Model
10. Context Data Model

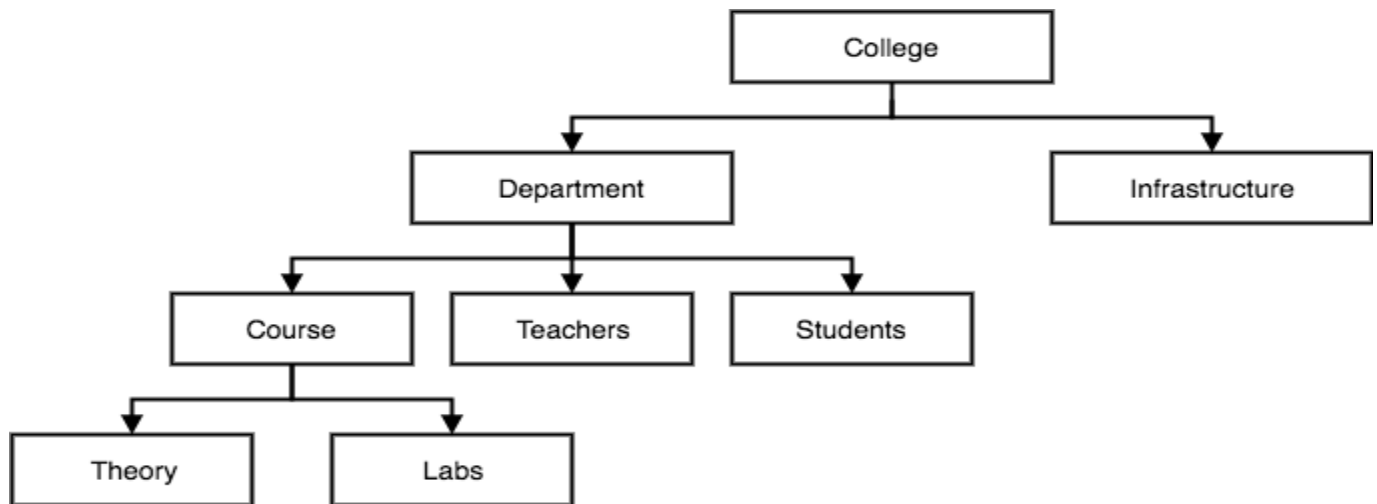
Hierarchical Model:

This database model organizes data into a tree-like-structure, with a single root, to which all the other data is linked. The hierarchy starts from the **Root** data, and expands like a tree, adding child nodes to the parent nodes.

In this model, a child node will only have a single parent node.

This model efficiently describes many real-world relationships like index of a book, recipes etc.

In hierarchical model, data is organized into tree-like structure with one one-to-many relationship between two different types of data, for example, one department can have many courses, many professors and off-course many students.

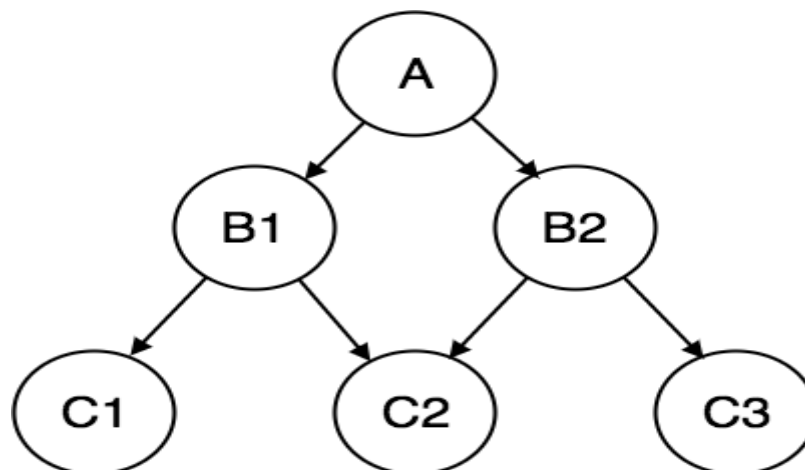


Network Model:

This is an extension of the Hierarchical model. In this model data is organized more like a graph, and are allowed to have more than one parent node.

In this database model data is more related as more relationships are established in this database model. Also, as the data is more related, hence accessing the data is also easier and fast. This database model was used to map many-to-many data relationships.

This was the most widely used database model, before Relational Model was introduced.



Entity-relationship Model:

In this database model, relationships are created by dividing object of interest into entity and its characteristics into attributes.

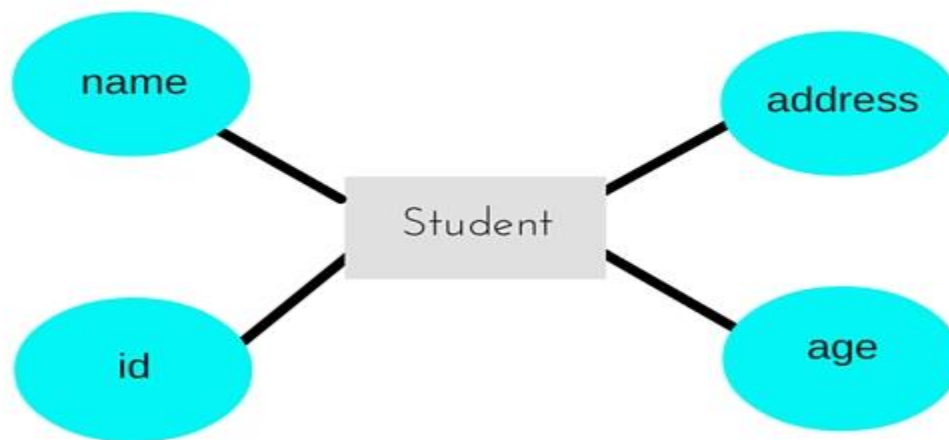
Different entities are related using relationships.

E-R Models are defined to represent the relationships into pictorial form to make it easier for different stakeholders to understand.

This model is good to design a database, which can then be turned into tables in relational model (explained below).

Let's take an example, if we have to design a School Database, then **Student** will be an **entity** with **attributes** name, age, address etc. As **Address** is generally complex, it can be another **entity** with **attributes** street name, pin code, city etc., and there will be a relationship between them.

Relationships can also be of different types. To learn about [E-R Diagrams](#) in details, click on the link.



Relational Model:

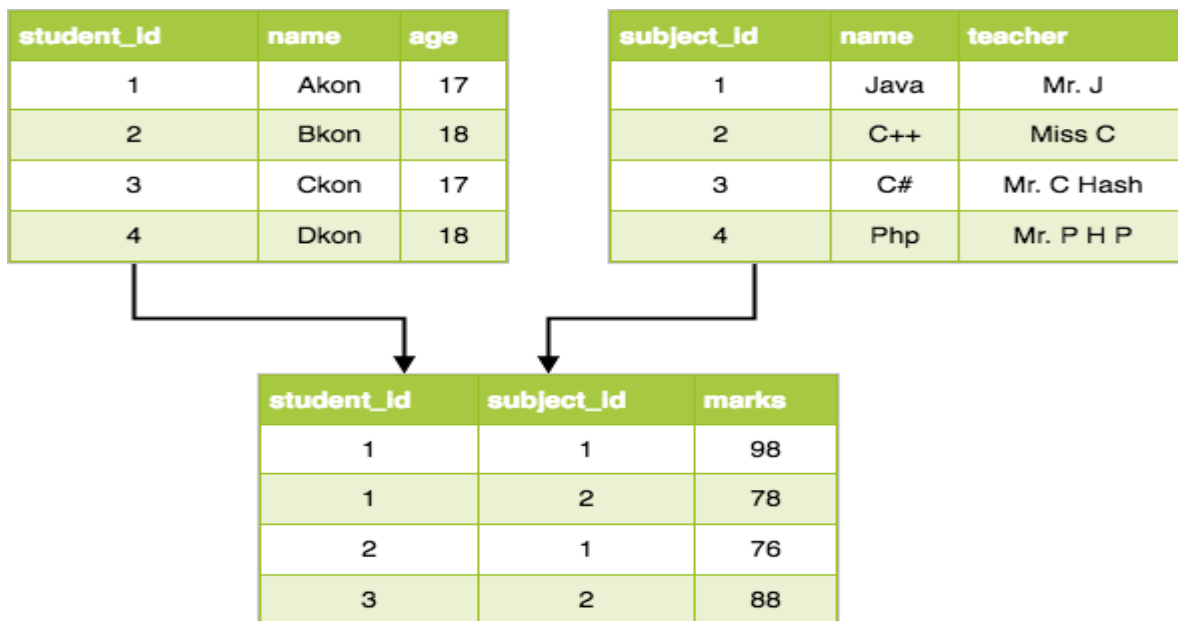
In this model, data is organized in two-dimensional **tables** and the relationship is maintained by storing a common field.

This model was introduced by E.F Codd in 1970, and since then it has been the most widely used database model, infact, we can say the only database model used around the world.

The basic structure of data in the relational model is tables. All the information related to a particular type is stored in rows of that table.

Hence, tables are also known as **relations** in relational model.

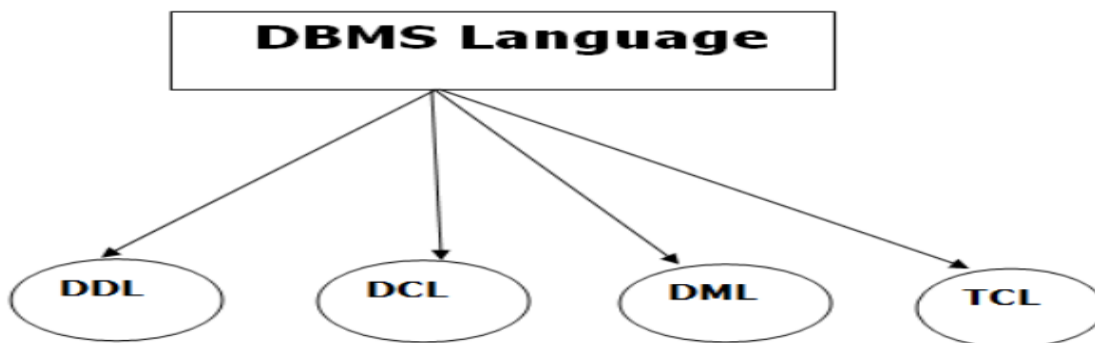
In the coming tutorials we will learn how to design tables, [normalize them](#) to reduce data redundancy and how to use [Structured Query language](#) to access data from tables.



6. Database Languages:

Database languages are used to read, update and store data in a database.

Types of Database Language



The following are the database languages

- Data Definition Language
- Data Manipulation Language
- Data Control Language
- Transaction Control Language

Data Definition Language:

The language is used to create database, tables, alter them, etc. With this, you can also rename the database, or drop them. It specifies the database schema.

The DDL statements include –

- **CREATE:** Create new database, table, etc.
- **ALTER:** Alter existing database, table, etc.
- **DROP:** Drop the database
- **RENAME:** Set a new name for the table.

Data Manipulation Language:

The language used to manipulate the database like inserting data, updating table, retrieving record from a table, etc. is known as Data Manipulation Language –

- **SELECT:** Retrieve data from the database
- **INSERT:** Insert data
- **UPDATE:** Update data
- **DELETE:** Delete all records

Data Control Language:

Grant privilege to a user using the GRANT statement. In the same way, revoke the privilege using the REVOKE statement. Both of these statements come under the Data Control Language (DCL).

- **GRANT:** Give privilege to access the database.
- **REVOKE:** Take back the privilege to access the database.

Transaction Control Language:

Manage transactions in the Database using the Transaction Control Language –

- **COMMIT:** Save the work.
- **SAVEPOINT:** Set a point in transaction to rollback later
- **ROLLBACK:** Restores since last commit